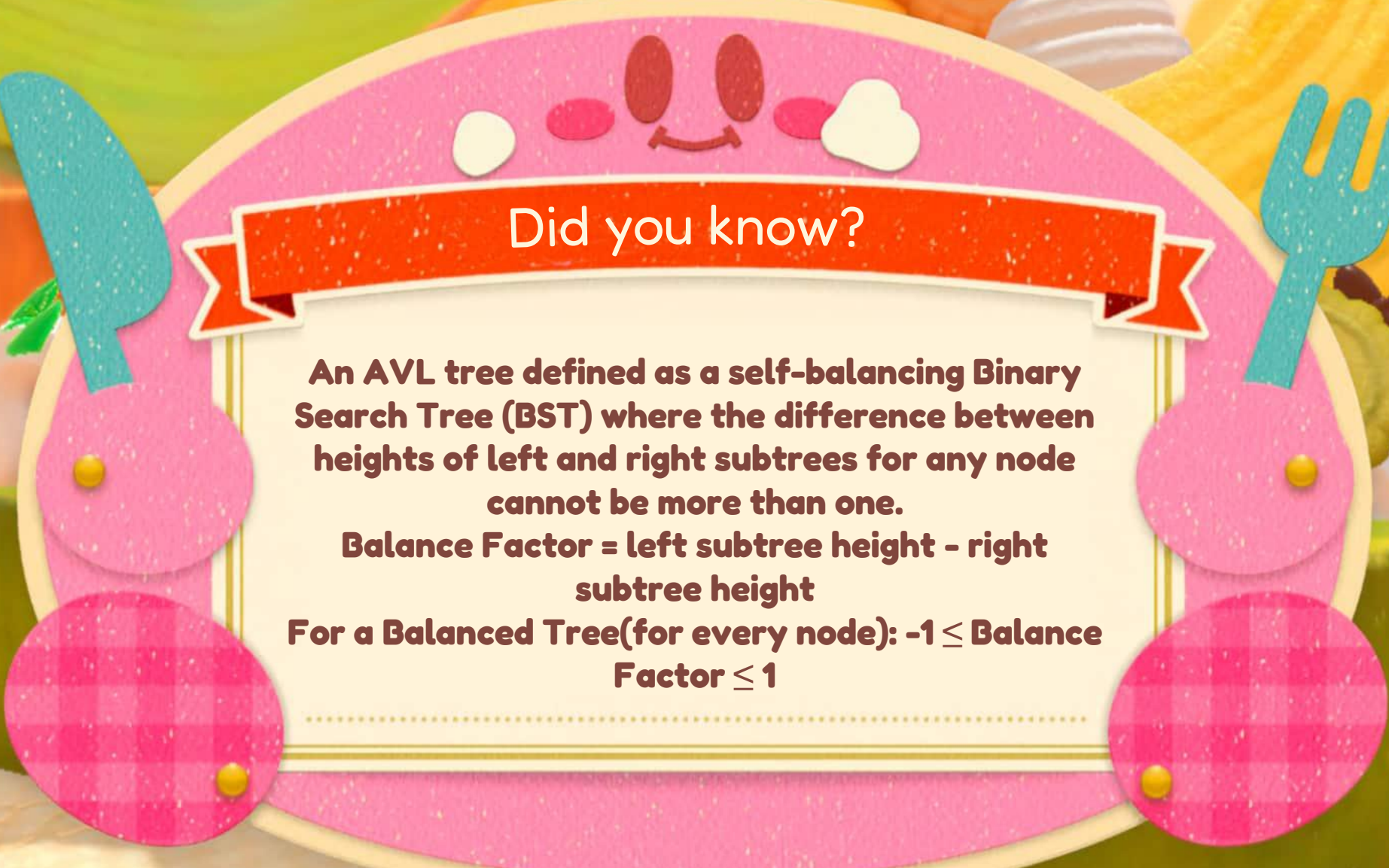


A pink Kirby character with large blue eyes and a small smile is peeking out from behind a large, bright yellow star. The star has a thick black outline and a slight gradient. The background is a solid pink color with a repeating pattern of small, faint Kirby characters in various poses and outfits. The top and bottom of the image are framed by a pink and white diagonal striped border.

AVL Trees

Self Balancing BST's



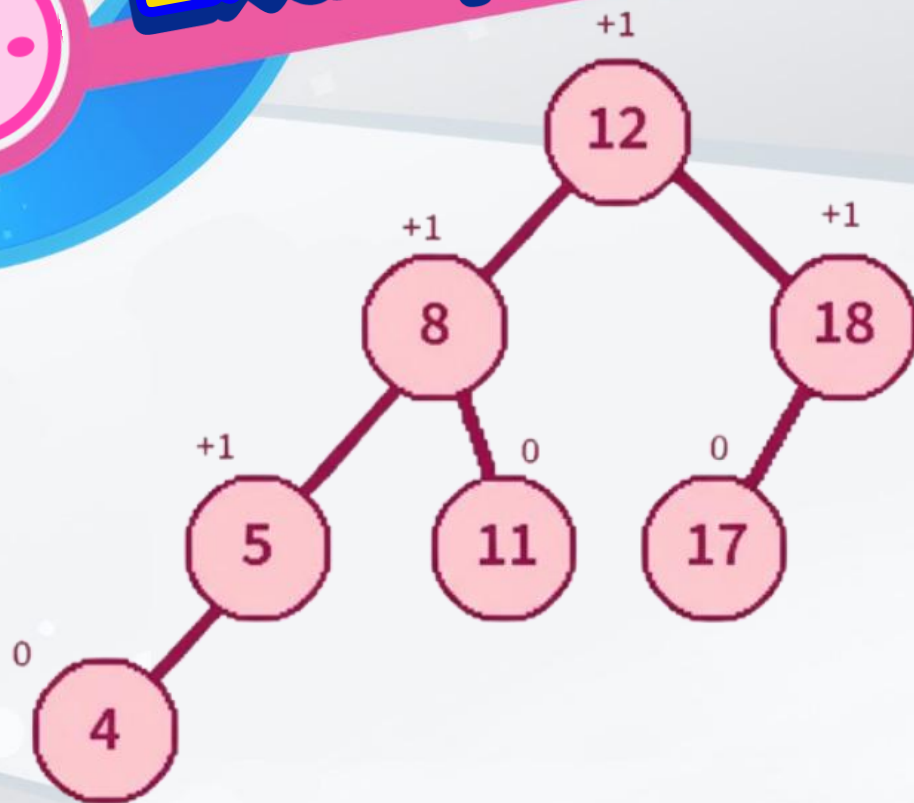
Did you know?

An AVL tree defined as a self-balancing Binary Search Tree (BST) where the difference between heights of left and right subtrees for any node cannot be more than one.

Balance Factor = left subtree height - right subtree height

For a Balanced Tree(for every node): $-1 \leq \text{Balance Factor} \leq 1$

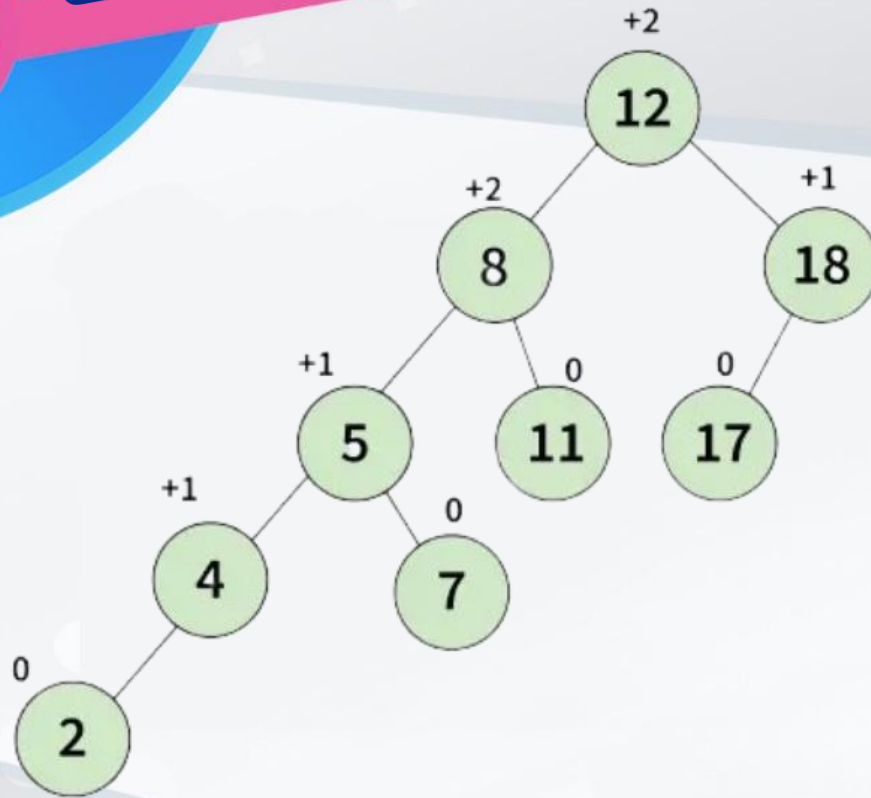
Example



Since all differences lie between -1 to +1, so the tree is an AVL tree.



Example



This BST which is not an AVL Tree because its not satisfying balancing factor condition.



5



Types Of Rotations



rotations are designed to restore balance

Right Left

Right Right

Left Left

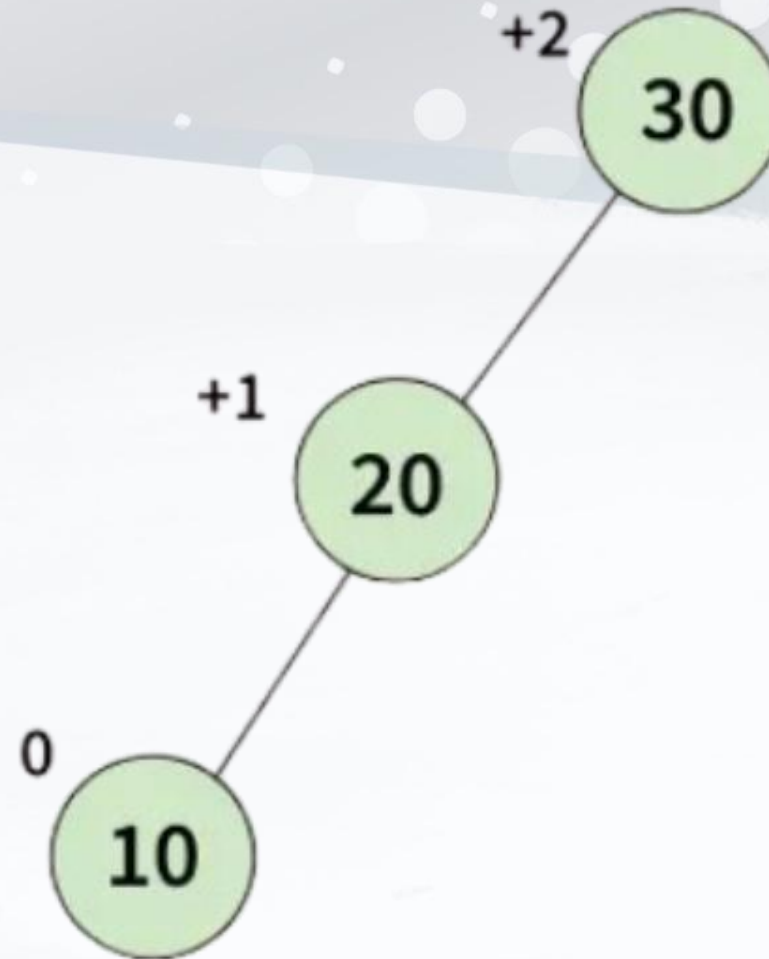
Left Right



Left Left

Node 30 has a balance factor of 2, $-1 < BF < 1$ is the acceptable range so we are going to perform left left rotation

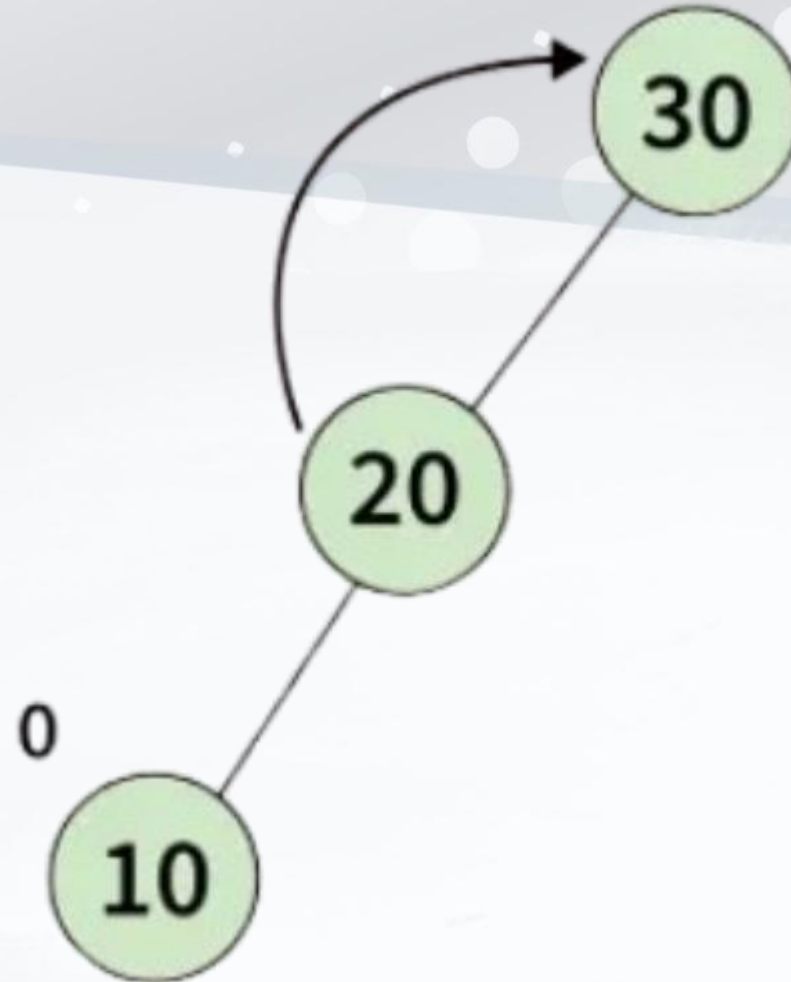
Left Left Rotation



Left Left

Left Left Rotation

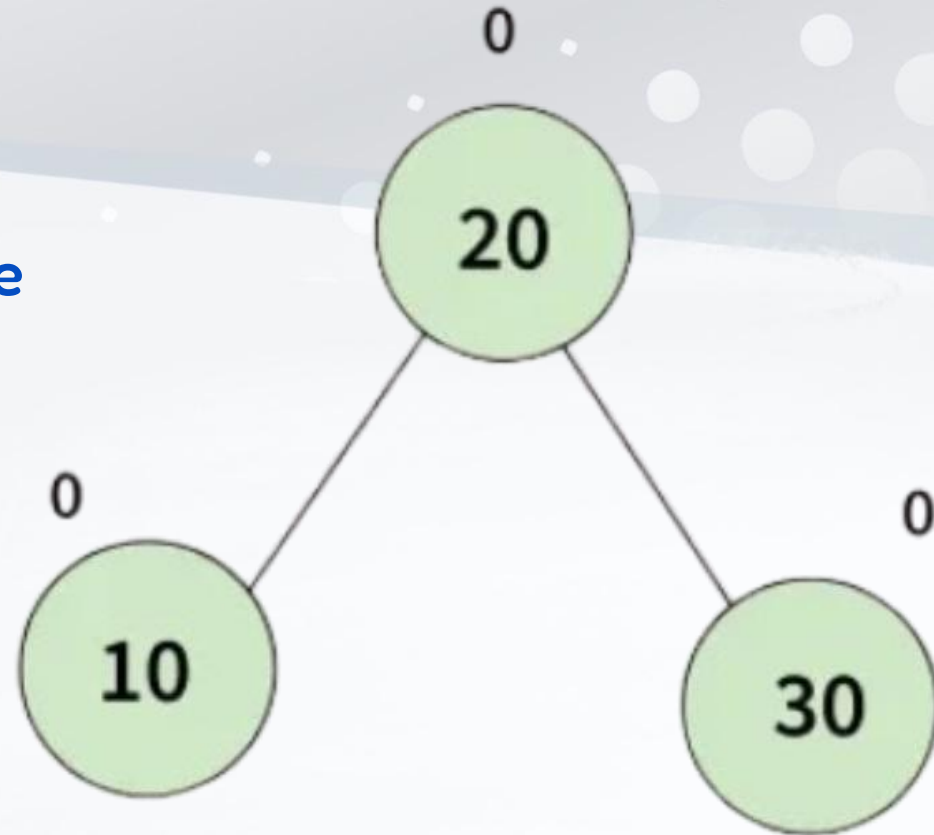
To achieve balance, a right rotation will be performed.



Left Left

Now, the tree is balanced as the balance factor isn't exceeding its range.

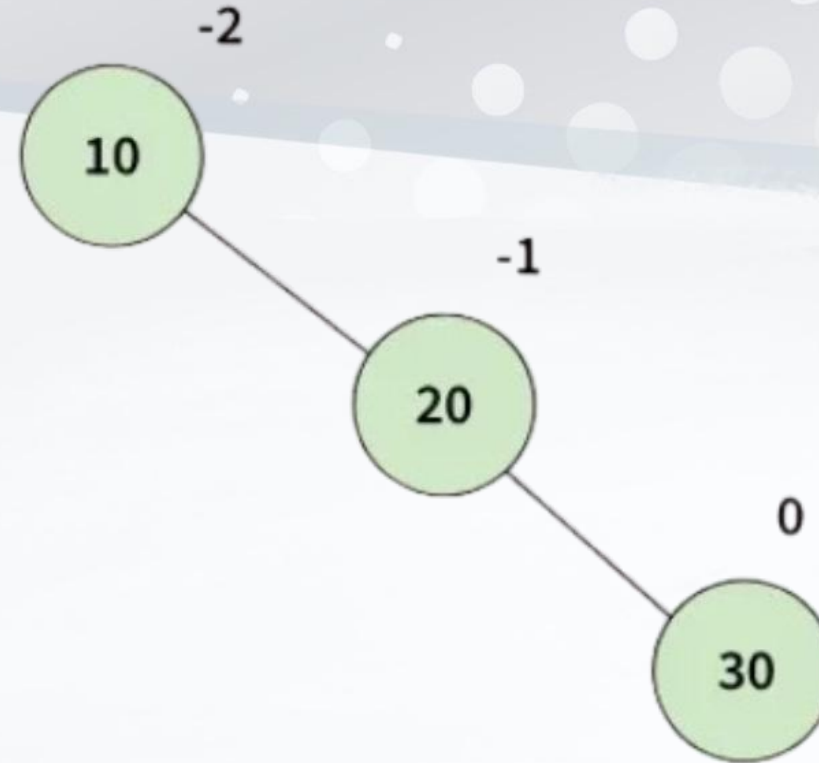
Left Left Rotation



Right Right

Right Right Rotation

Node 10 has a balance factor of -2, so performing right right rotation.

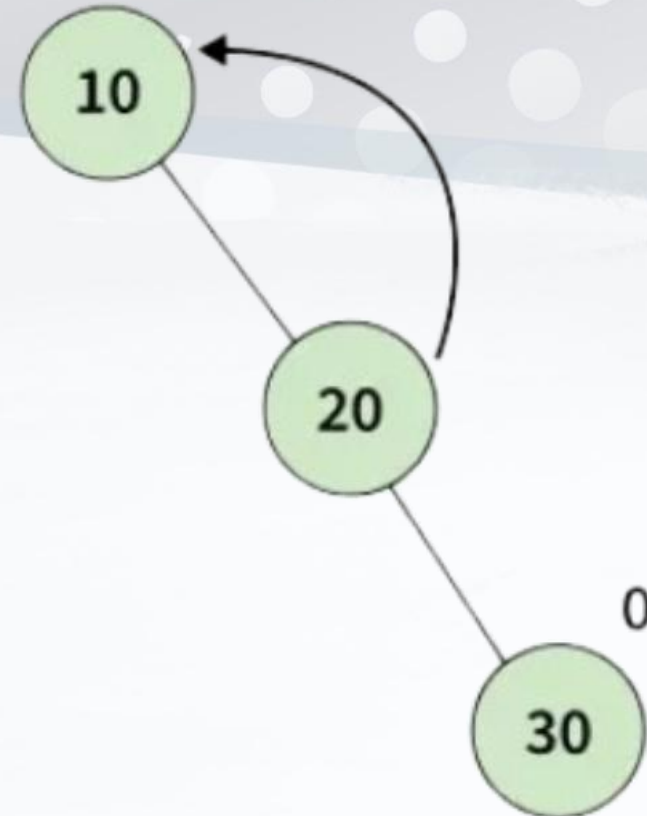


10

Right Right

Right Right Rotation

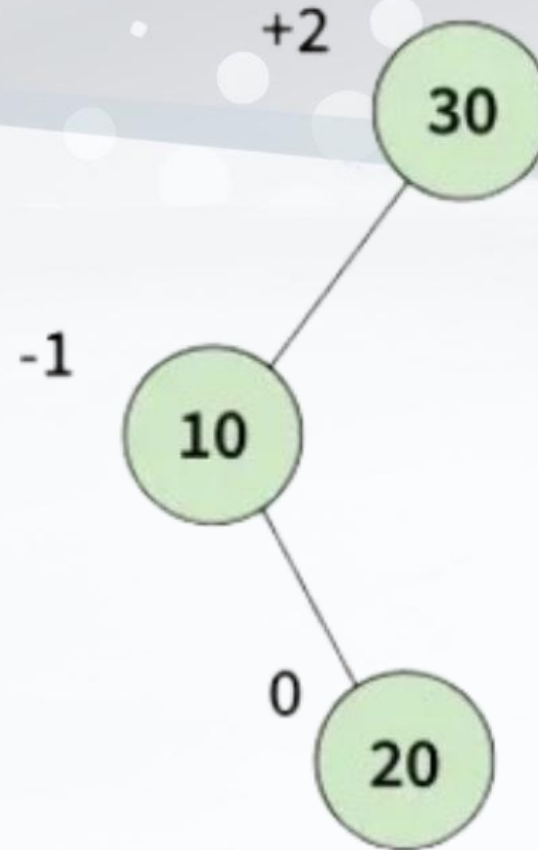
Left rotation is performed to balance it.



Left Right

Left Right Rotation

Occurs when a node is inserted into the right subtree of the left child, which disturbs the balance factor of an ancestor node, making it left-heavy.

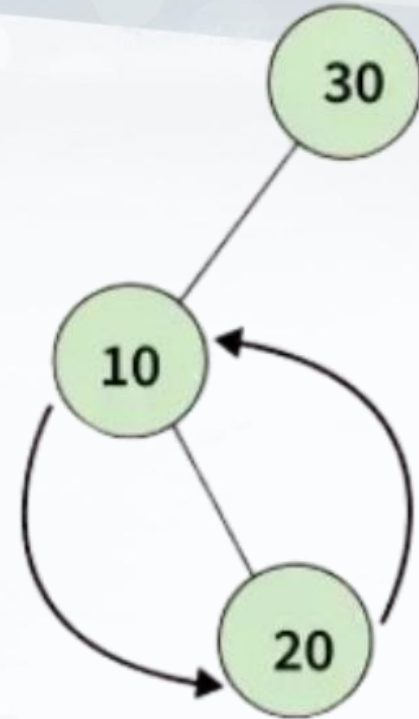


Left Right

Left Right Rotation

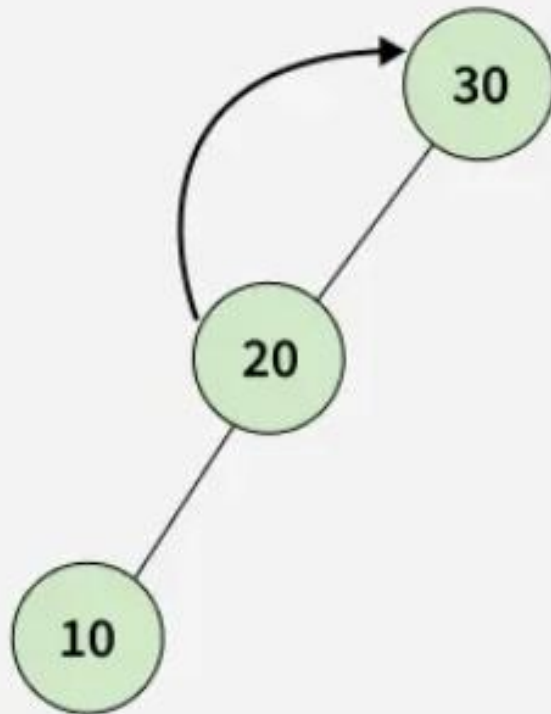
Performing left rotation

Fix: Perform a left rotation on the left child, followed by a right rotation on the node.



Left Right

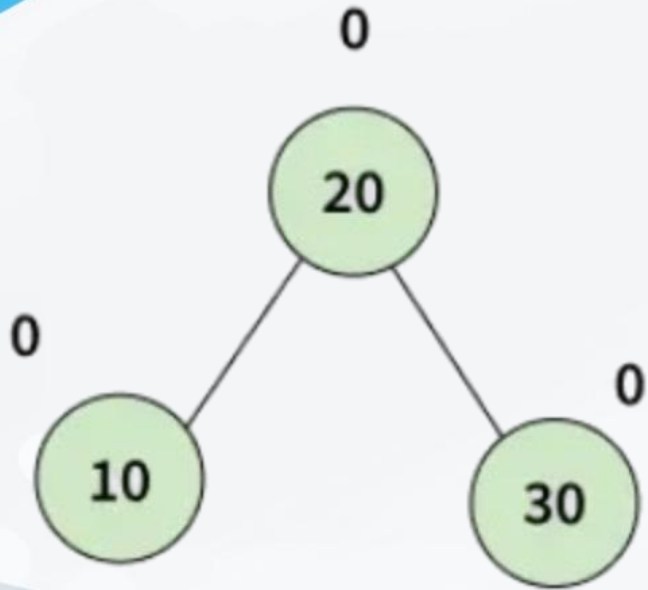
Left Right Rotation



To achieve balance in the tree, a right rotation is performed on nodes 20 and 30

Left Right

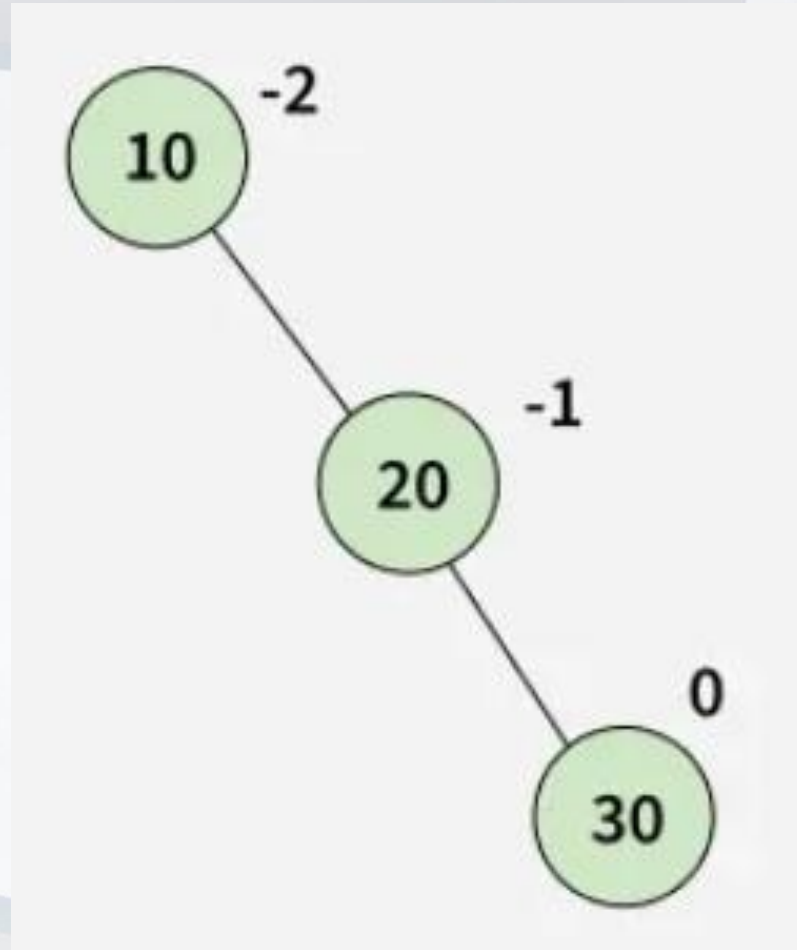
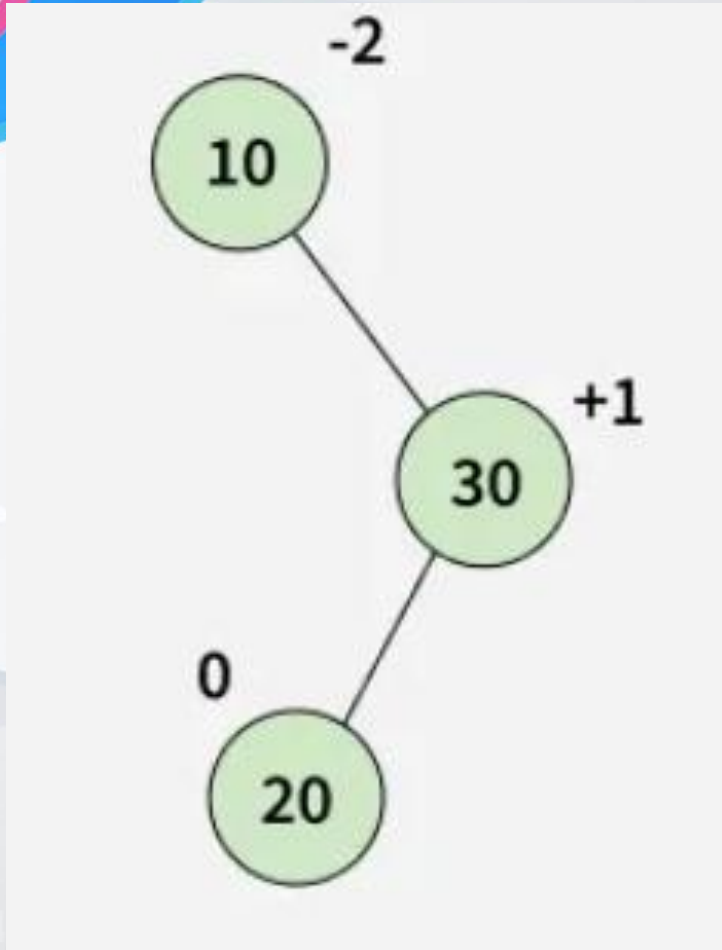
Left Right Rotation



The tree is balanced now, with all node balance factors within the valid range.

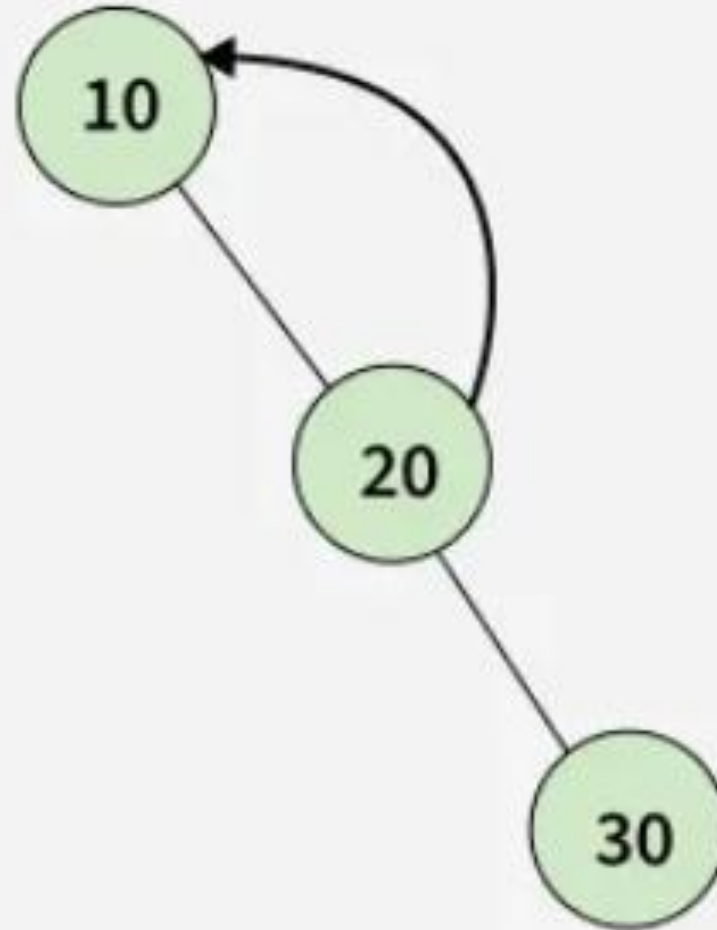
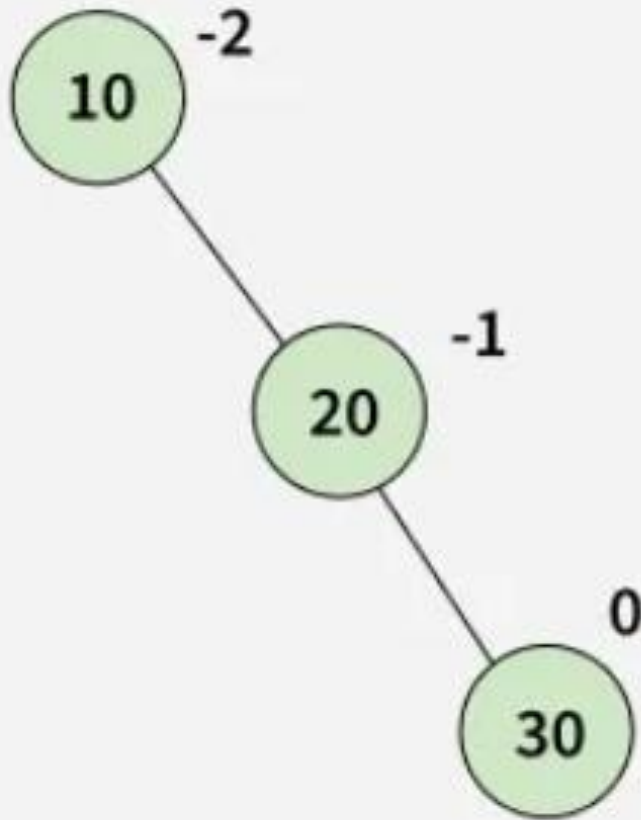
Right Left

Right Left Rotation



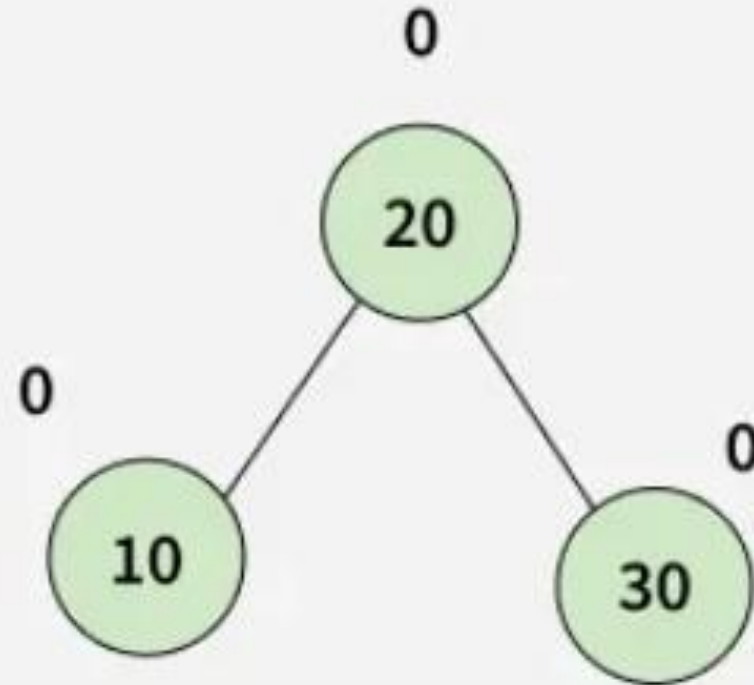
Right Left

Right Left Rotation



Right Left

Right Left Rotation





18



Advantages



$O(\log n)$ complexity

Fast search

Traversal in sorted order

Less Complex



DSA SEC E & F



Thank you 😊